MIL-M-455588

3 September 1974
SUPERSEDING
MIL-M-45558A
27 September 1968

MILITARY SPECIFICATION

MOISTURE STABILIZER, WELDING ELECTRODE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electrically heated moisture stabilizer used in protecting welding electrodes from absorbing excessive moisture.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

W-C-596	- Connector, Plug, Electrical; Connector, Receptacle, Electrical

Military

MIL-T-152	- Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible, and Extra-Flexible, 100 and 600 Volts)

FSC 3439

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

MIL-W-45562 - Welding and Soldering Equipment. Supplies and Accessories. Packaging of

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for inspection by Attributes

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-171 - Finishing of Metal and Wood Surfaces

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 499 - Electric Heating Appliances

UL 873 - Temperature-Indicating and Regulating Equipment

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

* Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the supplier shall furnish one complete moisture stabilizer for first article inspection. The first article may be either a preproduction model or an initial production item that conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The moisture stabilizer shall be of the manufacturer's latest design meeting the requirements specified herein.

3.2.1 Capacity. The moisture stabilizer shall be capable of storing 50 pounds of 18-inch welding electrodes in a vertical position.

* 3.2.2 Safety devices. The moisture stabilizer shall be fitted with safety devices or guards for all parts that present safety hazards. When required (see 6.2.1), special safety devices shall be as specified by the procuring activity.

3.2.3 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification to parts or basic design.

3.2.4 Material. All parts used in the fabrication shall be new and of the quality necessary to produce a moisture stabilizer to meet the requirements specified herein.

* 3.3 Construction. The moisture stabilizer shall be complete so that when connected to the specified source of power, it can be used for the operation specified herein. All parts subject to wear or replacement shall be accessible for adjustment and repair. Construction shall be free from any defects that will prevent the item from meeting the requirements specified herein.

3.3.1 Welding, brazing, and soldering. Welding, brazing, and soldering shall be employed only where these operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

3.3.2 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed. All fastening devices shall be made rust resistant by cadmium plating in accordance with finish 1.1.3.3 of MIL-STD-171.

3.3.3 Surfaces. All surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, or other harmful extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.3.4 Painting. Unless otherwise specified (see 6.2.1), painting and finishing of the moisture stabilizer shall be in accordance with the manufacturer's standard commercial practice.

3.3.5 Threads. All screw threads shall be in the inch system and shall conform to Handbook H28.

3.4 Performance and product characteristics.

3.4.1 Performance. Unless otherwise specified (see 6.2.1), a fixed operating temperature of not less than 200° Fahrenheit (F) shall be provided. When specified (see 6.2.1), an adjustable temperature range of at least 135°F to 220°F shall be provided. The moisture stabilizer shall heat contents by conduction, to the temperatures specified, in 2.5 hours or less and maintain the set operating temperature within ±8°F for a continuous period of 8 hours. At the end of this period the moisture stabilizer shall evidence no damage, distortion, warpage of the supporting structure, or impairment of function.

3.4.1.1 Heat loss. The average heat loss per square foot of internal surface shall not exceed 40 watts per hour at an ambient temperature of 70°F ±5°F.

3.4.2 Dielectric strength. The wiring of the electrical system, excluding controls, shall be capable of withstanding a voltage equal to two times the rated voltage plus 1,000 volts, root mean square (rms), at 60 hertz (Hz), for a period of one minute between the insulated circuits and between the insulated circuits and external metal parts.

3.5 Details of components. The moisture stabilizer shall consist of, but not be limited to, the following principal components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

* 3.5.1 Body. The body shall consist of an insulated double-wall container, with a cover attached by means of a metal piano-type hinge, or hinges, and securely closed by a latch. The walls, bottom, and cover shall be insulated. The latch shall be capable of being easily opened with a gloved hand. Two lifting handles, one on each side of the body, shall be furnished to afford easy portability, even with gloved hands. The body shall be rigid and sturdy in construction. Unless otherwise specified (see 6.2.1), the interior wall shall be metal, Manufacturers' Standard Gage No. 14 minimum thickness, and the exterior wall shall be metal, Manufacturers' Standard Gage No. 16 minimum thickness. The interior height shall be not less than 18.25 inches. The interior cross-section area shall be not less than 34 square inches. The cover shall provide a seal, when closed and secured, to assure a minimum of moisture absorption by the contents.

3.5.2 Heating element. The heating element shall operate from nominal 115-volt alternating current and direct current sources.

3.5.3 Control. Unless otherwise specified (see 6.2.1), the control shall be of a fixed temperature or, when specified (see 6.2.1), the control shall be of an adjustable temperature, sealed thermostat, or other control which interrupts line voltage to the heating element. The control shall be connected to a pilot light to indicate when the heating element is operating.

3.5.4 Power cord. The moisture stabilizer shall be provided with a heavy duty power cord, at least 10 feet long, that is in accordance with type CO-03HOF(3/18)0410 requirements specified in MIL-C-3432. The power cord shall be provided with a three-prong polarized plug conforming to style D of W-C-596, with mating three-to-two parallel prong adapter with grounding lead for use with a two-wire outlet.

3.5.5 Compartment dividers. The moisture stabilizer shall be furnished with metal compartment dividers which shall be easily removed from the container. When placed in the container, the dividers shall be capable of segregating four types of electrodes.

* 3.6 Electrical equipment. Electrical wiring and controls shall meet the requirements of UL 499 and UL 873.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

3.7 Fungus control. When specified (see 6.2.1), electrical components and circuit elements shall be treated for fungus resistance in accordance with the type II requirements of MIL-T-152, except that the statement of treatment is not required.

3.8 Identification marking. The moisture stabilizer shall be marked in accordance with MIL-STD-130, including serial number and Federal Stock Number.

* 3.9 Technical data. Data shall be furnished as specified (see 6.2.2).

3.10 Workmanship. Workmanship of the moisture stabilizer shall be of the quality prevailing among manufacturers normally producing moisture stabilizers of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection classification. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).

(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item noted in 3.7. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.3.1 and all tests under 4.3.2, 4.3.3, and 4.3.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to each lot of completed items being submitted for acceptance under the contract. The inspection shall consist of the following:

(a) Examination (see 4.3.1).

(b) Performance test (see 4.3.2).

(c) Inspection of preparation for delivery (see 4.4).

4.2.2.1 Sampling. Sampling for lot acceptance shall be in accordance with MIL-STD-105 at the inspection and acceptable quality levels specified by the procuring activity. Samples for the test specified shall be selected at random from the samples that passed the examination.

4.3 Inspection procedure.

4.3.1 Examination. The moisture stabilizer and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of the specification. Any deviation from these requirements shall be cause for rejection.

4.3.2 Performance test. The moisture stabilizer shall have 50 pounds of welding electrodes placed within it (see 3.2.1). With the cover closed and secured, the temperature shall be raised from ambient room temperature, 70°F ±5°F, to the operating temperature (see 3.4.1). When an adjustable temperature range is provided this test shall be conducted with the temperature control set at the lowest temperature setting and again with the temperature control set at the highest temperature setting. The required operating temperature, measured at the top center of the load, shall be attained in 2.5 hours or less. For first article test only, the required operating temperature shall be maintained within ±8°F for a continuous period of 8 hours. Failure, or excessive time, to attain operating temperature, variation from temperature control specified, indication of damage or distortion, or warpage of the supporting structure shall be cause for rejection (see 3.4.1).

4.3.3 Heat loss test. The heat loss test shall run concurrently with 4.3.2 after the operating temperature is attained and stabilized within prescribed limits for at least 2 hours in an average room temperature of 70°F ±5°F. The average power required to maintain the stabilized condition for a period of not less than 5 hours shall not exceed 40 watts per hour per square foot of internal surface. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

average heat loss in excess of the requirements specified in 3.4.1.1 shall be cause for rejection.

4.3.4 Dielectric test. The wiring of the electrical system, excluding controls, shall be subjected to a voltage equal to two times the rated voltage plus 1,000 volts rms, at 60 Hz, for a period of one minute between the insulated circuits and also between the insulated circuits and the external metal parts. On conclusion of test, any evidence of damage shall be cause for rejection (see 3.4.2).

4.4 Inspection of preparation for delivery. Preservation-packaging, packing, and marking for shipment and storage shall be in inspected to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation-packaging, packing, and marking. Unless otherwise specified (see 6.2.1), preservation-packaging, packing, and marking shall be in accordance with MIL-W-45562. The levels of preservation-packaging and packing shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The moisture stabilizer covered by this specification will be primarily used for protecting welding electrodes from excessive moisture absorption in field and shop maintenance areas.

* 6.2 Ordering data. Procurement documents should specify:

* 6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) First article, when required (see 3.1).

(c) Special safety devices, when required (see 3.2.2).

(d) Painting, if different (see 3.3.4).

(e) Constant temperature, if different (see 3.4.1).

(f) Adjustable temperature, when required (see 3.4.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.

(g) Interior and exterior wall material, if different (see 3.5.1).

(h) Fixed control, if different (see 3.5.3).

(i) Adjustable control, when required (see 3.5.3).

(j) Fungus control, when required (see 3.7).

(k) First article inspection, if different (see 4.2.1).

(l) Quality conformance inspection, if different (see 4.2.2).

(m) Preservation-packaging, packing, and marking, if different (see 5.1).

(n) Selection of applicable level of preservation-packaging, and packing (see 5.1).

* 6.2.2 Contract data requirements. Required technical data, such as operator's manuals, parts lists, drawings, schematics, wiring instructions, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.9).

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
Navy - None
Air Force - 84

Review activities:
Army - EL
Navy - None
Air Force - None
DSA - GS

User Activities:
Army - SM
Navy - None
Air Force - None

Preparing activity:
Army - WC

Project No. 3439-0199

☆ U.S. GOVERNMENT PRINTING OFFICE: 1974-603-118/2113

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:24Z
Check the source to verify that this is the current version before use.